

Northern University of Business and Technology, Khulna

Department of Computer Science and Engineering (CSE)



Course Title: Software Development Project

Course Code: CSE 4104

Assignment Title: Week 05: UI/UX Design and Development Planning

Team Name: CSE4104-7A-T04

Section: 7A

Project Title: AI Study Assistant Platform

Team Leader:

Name: Utpal Saha

ID: 11230121085

Team Members:

1. Utpal Saha— ID: 11230121085
2. Prianka Mondal— ID:11230121089
3. Anuja Das — ID: 11230121096
4. Nazmul Hasan Emon — ID: 11230121098

Submitted To:

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Submitted By:

Team: CSE4104-7A-T04

Date:22 June 2026

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1. TEAM INFORMATION

Team Name: CSE4104-7A-T04

Project Title: AI Study Assistant Platform

Team Leader:

- Name: Utpal Saha
- ID: 11230121085

Members:

1. Utpal Saha— ID: 11230121085
2. Prianka Mondal— ID:11230121089
3. Anuja Das — ID: 11230121096
4. Nazmul Hasan Emon — ID: 11230121098

GitHub Repository Link: <https://github.com/UtpalSaha21/cse4104-7a-t04-AI-Study-Assistant-Platform>

2. PROJECT DESCRIPTION

The AI Study Assistant Platform is a web-based intelligent learning system designed to assist students in their academic activities. The platform provides AI-powered features such as an interactive chatbot, note summarization, quiz generation, task management, and user profile management. By integrating artificial intelligence, the system aims to enhance students' learning experience and improve productivity through personalized academic support.

The platform will be developed using React.js for the frontend, Node.js and Express.js for the backend, MongoDB as the database, and Gemini API for AI functionalities.

3. UI/WIREFRAME SCREENSHOTS

Registration Page



Start Your Academic Journey

Join thousands of students mastering their subjects with AI-powered intuition.

 **AI Study Assistant**

Create Your Account

Step into the future of learning today.

Full Name

Email Address

Password

Create Account

or

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Create Your Account

Step into the future of learning today.

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Enter your full name

Email Address

example@scholarai.edu

Password

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Create Account

or



Sign up with Google

Already have an account? [Log in](#)

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Login Page



Master Your Subjects with Clarity.

Experience a scholarly workspace where AI meets intuition, designed for deep focus and academic growth.

[Adaptive Learning](#)[Deep Focus](#)



Welcome Back

Continue to your journey with mastery

Email Address

student@university.edu

Password

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[Forget password?](#)

Sign In

or

 Sign up with Google

Don't have an account? [Sign up](#)

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Welcome Back

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Email Address

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Password

[Forget password?](#)

.....

Sign In

or



Sign up with Google

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Dashboard

 **AI Study Assistant**
Academic Mentor

[Dashboard](#)

[Ask AI](#)

[Summarizer](#)

[Quizzes](#)

[Planner](#)

[Setting](#)

[Support](#)

 **Alex Johnson**
Student 

Good afternoon, Alex!

What are we mastering today? Your Physics exam is in 3 days—let's keep the momentum going.

[Resume Last Lesson](#)

Quick Actions

**New Chat**
Ask anything about your syllabus

**Upload Doc**
Get summaries from PDFs or images

**Create Quiz**
Generate mock tests instantly

Recent Summaries

[View All](#)

 **Quantum_Mechanics_Lecture_04.pdf**
Summarized 2 hours ago • 12 key concepts identified

 **Modern_World_History_Notes.docx**
Summarized Yesterday • 8 key dates highlighted

DAILY PROGRESS

75% COMPLETE

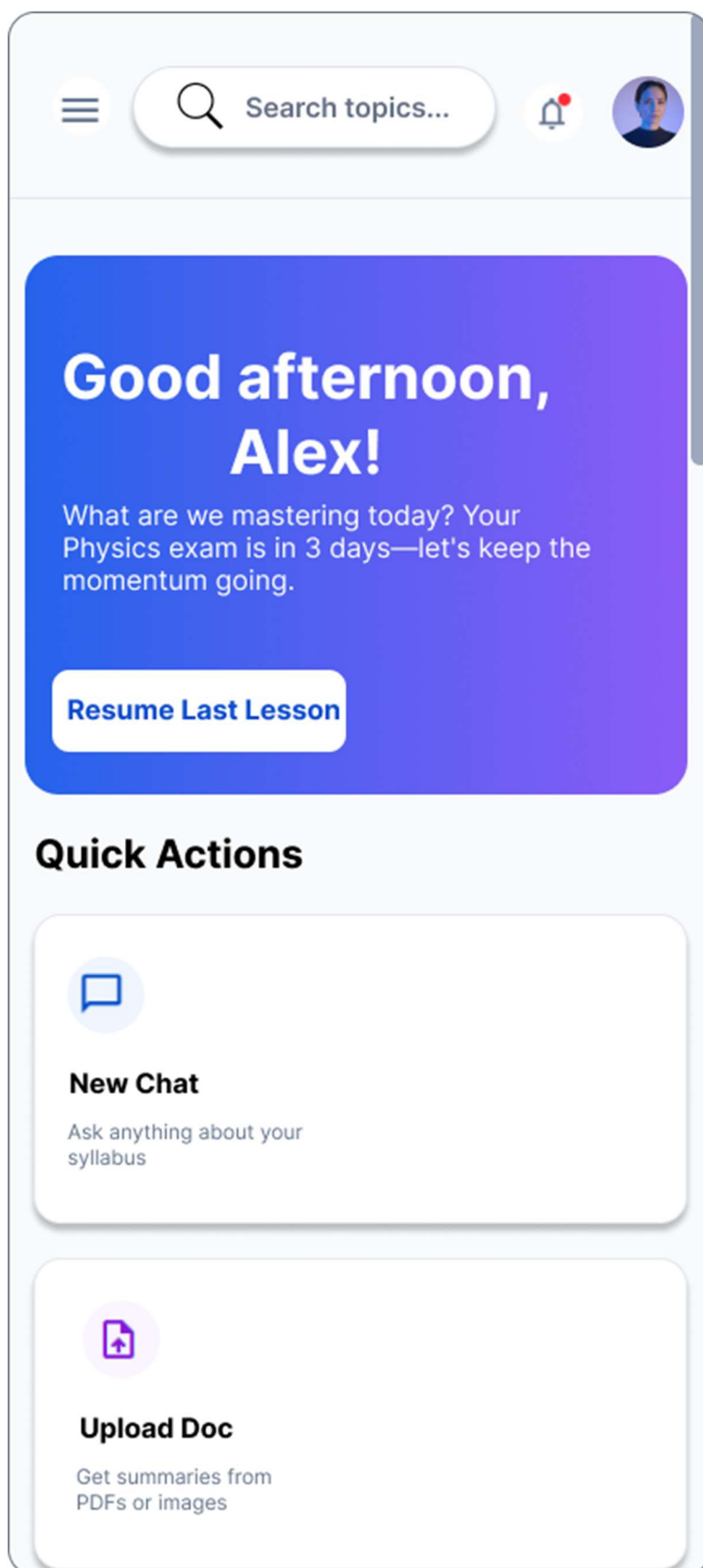
Almost there! Only 45 minutes left to hit your daily goal.

[STREAK: 12 DAYS](#)

Upcoming Tasks

 **History Quiz: Cold War Era**
Today, 2:00 PM

 **Submit Physics Summary**
Tomorrow, 10:00 AM



Profile

**AI Study Assistant**
Academic Mentor

Dashboard

Ask AI

Summarizer

Quizzes

Planner

Setting

Support

Log Out

Search topics...



Alex Johnson
Student





Alex Johnson

Student

Physics enthusiast dedicated to mastering quantum mechanics and helping peers bridge the gap between theory and practice.

Edit Profile

Share Portfolio

Academic Profile

Major
Physics

University
Stanford University

Current GPA
3.8 / 4.0

Account Settings

Manage your digital workspace...

Account Security

Password, 2FA, and login activity

Notification Preferences

Email alerts and in-app study reminders

Subscription Plan

Current: Scholar Pro - Next billing Sep 12

Connected Apps

Google Drive, Notion, and Slack integrations

Deactivate Account

Permanently remove all your data and access.

Delete Account

[9]

AI Chat Page

AI Study Assistant
Academic Mentor

Dashboard

Ask AI

Summarizer

Quizzes

Planner

 Alex Johnson
Student

 Setting

 Support

 Log Out

 + New Chat

RECENTS

Quantum Mechanics Intro ...

French Verb Conjugation ...

Calculus Derivatives ...

AI Study Assistant

Current SessionHistory



Session Started

Hello Alex! I'm ready to help you with your Calculus and Physics homework today. What would you like to explore first?

Can you help me calculate the first derivative of the function $f(x) = 3x^3 + 5x^2 - 12$ in Python? Also, please explain the power rule briefly.



Of course! To find the derivative using the Power Rule, you multiply the coefficient by the exponent and then subtract one from the exponent ($nx^{(n-1)}$). Constants like -12 become zero.

```
import sympy as sp

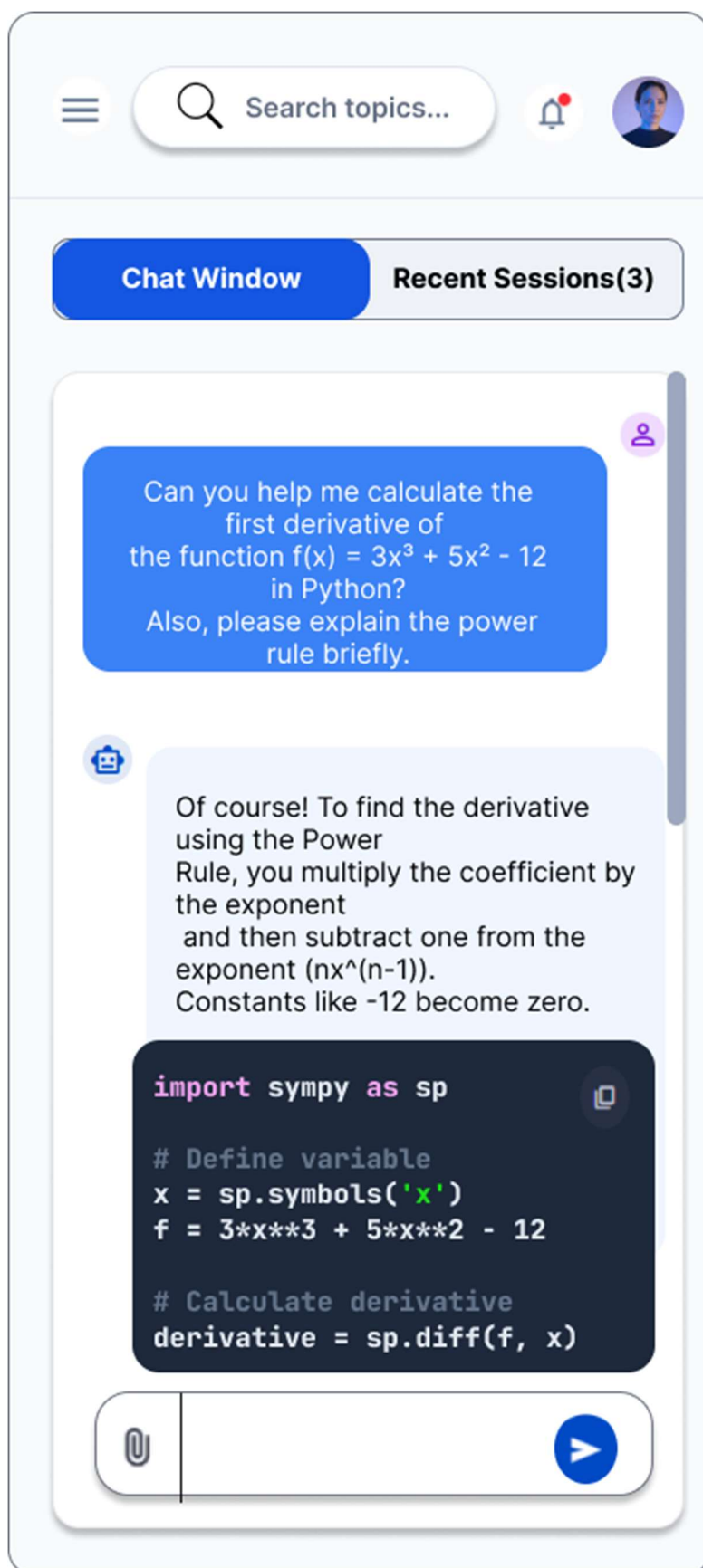
# Define variable
x = sp.symbols('x')
f = 3*x**3 + 5*x**2 - 12

# Calculate derivative
derivative = sp.diff(f, x)
print(derivative) # Result: 9*x**2 + 10*x
```



ScholarAI can make mistakes. Verify important academic information

[10]



Quiz Page

**AI Study Assistant**
Academic Mentor

Dashboard

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Search topics...

Alex Johnson
Student

Quiz & Test Center

Challenge yourself with AI-generated assessments tailored to your current study curriculum.

Drag and Drop your PDF/Notes here
Supports PDF, DOCX, and TXT files up to 50MB

Browse Files

Subject

e.g MolecularBiology

v

Number of Questions

5

10

15

20

Generate Quiz

Recent Quiz Results

View All

Subject & Topic	Date Attempted	Score
Keynesian Models & Multipliers Economics	Oct 24, 2023	85%

Molecular Biology Mastery
Question 3 of 10

TIME REMAINING
12:45

What is the primary function of the mitochondria in a cell?

A. Protein synthesis and folding

B. Generation of chemical energy (ATP)

C. Storage of genetic information

D. Waste removal and detoxification

Previous

Next Question →

Explanation

Mitochondria are known as the "powerhouse of the cell" because they generate ATP through cellular respiration. Protein synthesis occurs in ribosomes, genetic information is stored in the nucleus, and waste removal is done by lysosomes.

Search topics...



**Drag and Drop your
PDF/Notes here**

Supports PDF, DOCX, and TXT files up to
50MB

Browse Files

Subject

e.g MolecularBiology

v

Number of Questions

5

10

15

20

Generate Quiz

Recent Quiz Results

View All

Subject & Topic	Date Attempted	Score
Keynesian Models & Multipliers Economics	Oct 24, 2023	85%

Summary Page



AI Study Assistant
Academic Mentor

Dashboard

Ask AI

Summarizer

Quizzes

Planner

Setting

Support

Log Out

Search topics...



Alex Johnson Student 

Summarizer

Transform dense academic papers and lecture notes into concise, actionable study materials instantly using advanced AI analysis.



Drag and Drop your PDF/Notes here

Supports PDF, DOCX, and TXT files up to 50MB

Browse Files

Document Title

e.g MolecularBiology

Paste raw notes text or essay

Paste chapter excerpts.wikipedia paragraph etc.....

Summarize

Key Highlights

Content Analysis

Quantum Mechanics

Fundamental Concepts

The core thesis revolves around the principle of **quantum superposition**, where particles exist in multiple states simultaneously until measured.

Key Observations

- Observation of interference patterns in double-slit experiments.
- Correlation between observer interaction and wave function collapse.
- Non-local entanglement across significant spatial distances.

Contextual Background

Research indicates that classical physics fails to explain microscopic phenomena, necessitating a probabilistic framework.

Add to Library

Share Notes

Recents Uploads

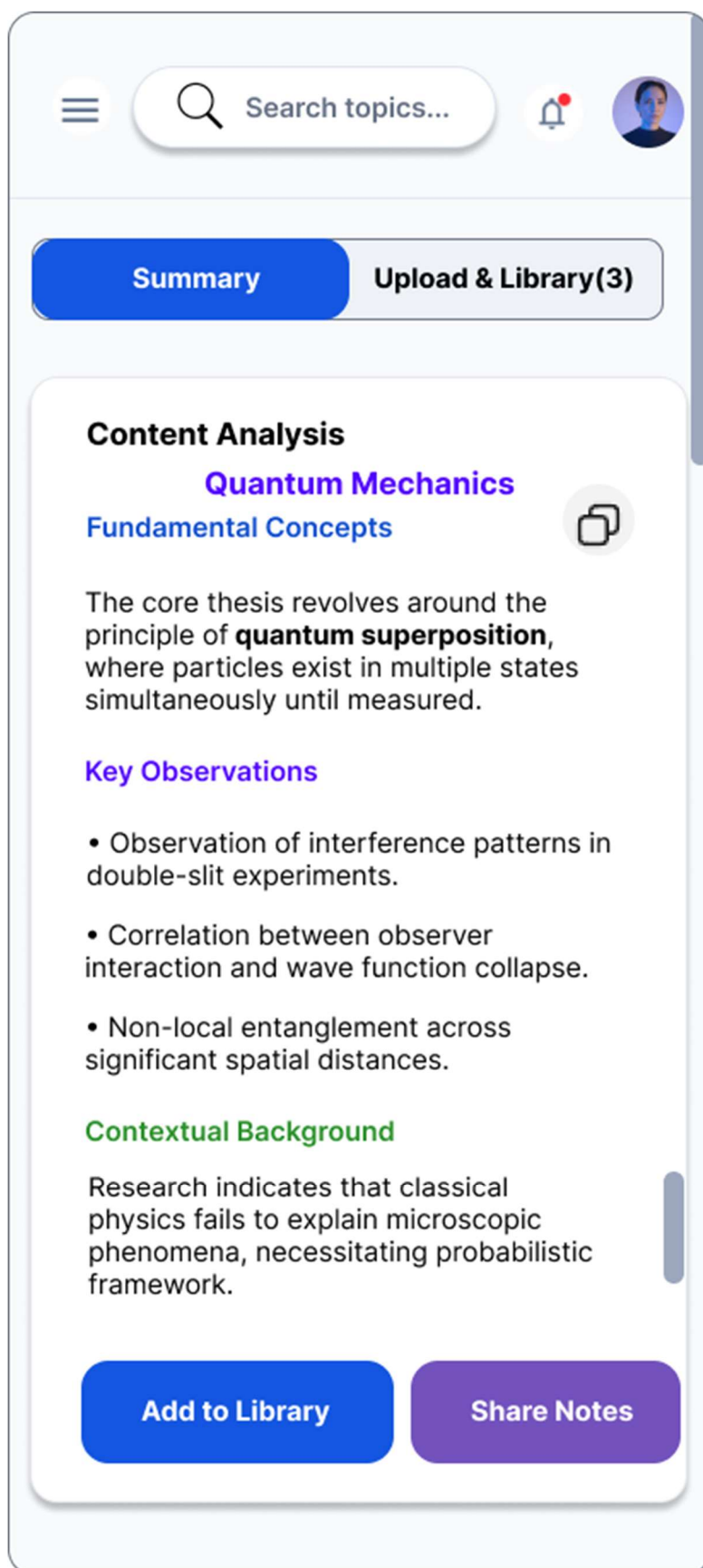
View All



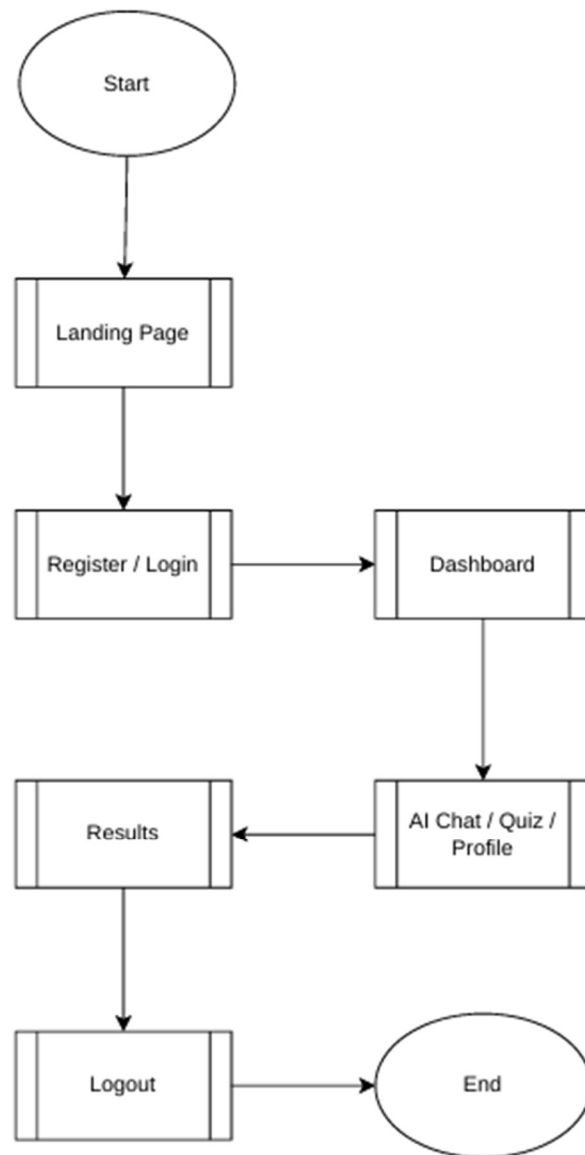
Quantum_Mechanics_Lecture_04.pdf

Summarized 2 hours ago • 12 key concepts identified

[14]



4. USER FLOW DIAGRAM



5. DEVELOPMENT ROADMAP

i. Project Timeline:

Week	Phase	Activities
Week 1	Project Idea Selection	Brainstorm project ideas, select AI Study Assistant Platform
Week 2	Proposal Preparation	Project proposal, team information sheet, GitHub repository setup
Week 3	SRS Preparation	Create SRS document, define requirements, prepare initial diagrams
Week 4	System Design	Create Architecture Diagram, ER Diagram, Use Case Diagram, API Structure
Week 5	UI/UX Design	Design Login Page, Dashboard, AI Chat Interface, Profile Page using Figma
Week 6	Frontend Setup	Setup React project structure, routing, reusable components
Week 7	Backend Setup	Setup Node.js, Express.js server, MongoDB connection
Week 8	Authentication Module	User registration, login, JWT authentication, profile management
Week 9	Dashboard Development	Create dashboard UI and integrate user statistics
Week 10	AI Chatbot Integration	Connect Gemini API, implement AI chat functionality

Week	Phase	Activities
Week 11	Notes Summarization Module	Develop note upload and AI-powered summary generation
Week 12	Quiz Generation Module	Implement AI-based quiz generation and quiz history
Week 13	Testing & Bug Fixing	System testing, debugging, performance improvements
Week 14	Deployment & Final Documentation	Deploy project, finalize documentation, prepare presentation
Week 15	Final Presentation & Submission	Demonstrate project and submit final deliverables

ii. Development Milestones:

Milestone 1: Planning Phase

- Project Proposal
- Team Formation
- GitHub Repository Setup
- Requirement Analysis

Milestone 2: Design Phase

- SRS Document
- Use Case Diagram
- ER Diagram
- System Architecture Diagram
- API Design

- UI/UX Design

Milestone 3: Development Phase

- Frontend Development
- Backend Development
- Database Integration
- Authentication System
- AI Feature Integration

Milestone 4: Testing Phase

- Functional Testing
- UI Testing
- API Testing
- Bug Fixing

Milestone 5: Deployment Phase

- System Deployment
- Final Documentation
- Project Demonstration

iii. Expected Outcome:

- AI-powered academic assistance
- Intelligent chatbot support
- AI-generated quizzes
- Note summarization
- User profile management
- Responsive web interface
- Secure authentication system

6. TEAM TASK DISTRIBUTION

Team Member	Responsibility
Team Leader	Project Coordination, Documentation, Testing and Debugging
Member 2	AI Integration using Gemini API
Member 3	Backend Development (Node.js, Express.js), Authentication APIs, Database Management (MongoDB)
Member 4	Frontend Development (React.js), User Interface Implementation

Detailed Responsibilities

Team Leader

- Coordinate project activities and monitor progress.
- Prepare project documentation and reports.
- Perform system testing and debugging.

Member 2

- Integrate Gemini API into the system.
- Develop AI-powered functionalities such as chatbot interaction, note summarization, and quiz generation.

Member 3

- Develop backend services using Node.js and Express.js.
- Implement authentication and authorization APIs.
- Design and manage MongoDB collections and relationships.

Member 4

- Develop responsive user interfaces using React.js.
- Implement UI designs and connect frontend components with backend APIs.

7. TECHNOLOGY STACK

Frontend

- React.js
- HTML5
- CSS3
- JavaScript

Backend

- Node.js
- Express.js

Database

- MongoDB Atlas

AI Integration

- Gemini API

Version Control

- Git & GitHub